



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Course Code: CSA-1522

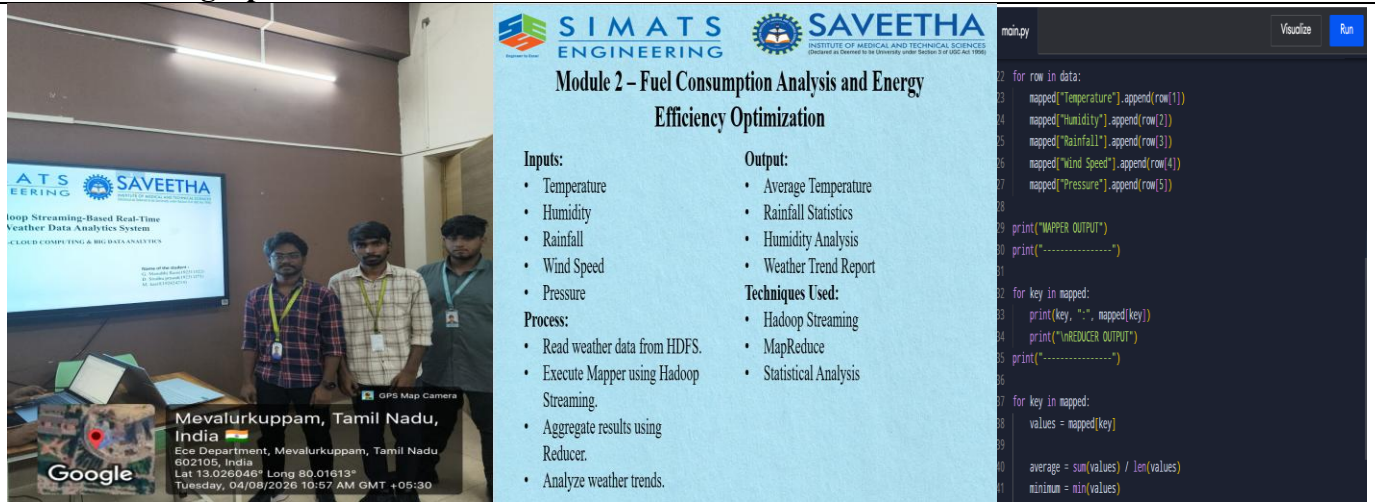
Slot: C

Course Name: Cloud Computing & Big Data Analytics

Course Faculty: Dr.P.Rajaram

Project Title: Hadoop Streaming-Based Real-Time Weather Data Analytics System

Module Photographs:



The photograph shows a presentation slide for Module 2, titled "Module 2 - Fuel Consumption Analysis and Energy Efficiency Optimization". The slide is divided into several sections: Inputs, Output, Process, and Techniques Used. The Inputs section lists Temperature, Humidity, Rainfall, Wind Speed, Pressure, and Process. The Output section lists Average Temperature, Rainfall Statistics, Humidity Analysis, Weather Trend Report, and Techniques Used. The Process section lists Read weather data from HDFS, Execute Mapper using Hadoop Streaming, Aggregate results using Reducer, and Analyze weather trends. The Techniques Used section lists Hadoop Streaming, MapReduce, and Statistical Analysis. The slide also includes a Google Map of Mevalurkuppam, Tamil Nadu, India, with coordinates Lat 13.020046° Long 80.01613° and a timestamp of Tuesday, 04/09/2020 10:17:AM GMT +05:30. The slide is presented by SIMATS ENGINEERING and SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES.

Project Description:

The project “Hadoop Streaming-Based Real-Time Weather Data Analytics System” aims to provide an efficient approach for processing and analyzing large-scale weather data for effective monitoring and decision-making. Module 2 focuses on Weather Data Analysis Using Hadoop Streaming, which is the core processing stage responsible for converting stored weather data into meaningful statistical information and identifying important weather trends. In this module, weather parameters such as temperature, humidity, rainfall, wind speed, and atmospheric pressure are retrieved from HDFS and processed using the Hadoop Streaming framework. The Mapper processes individual weather records and extracts the required information, while the Reducer groups and aggregates the processed data to generate useful results. Statistical analysis is performed to determine values such as average temperature, rainfall statistics, and humidity levels, while trend analysis helps identify variations and patterns in weather conditions. The final outputs of this module include average temperature, rainfall statistics, humidity analysis, and weather trend reports, which are subsequently used by the monitoring system for displaying analyzed weather information and supporting timely decision-making.

Student Signature

Guide Signature